Raaghav Thirumaligai

Prototyping Designer ~ Mechanical Engineer ~ Controls PhD Student

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Profile

Prototyping designer and mechanical engineer exploring how new hardware can create new interactions. Builds physical mockups, motion studies, and functional prototypes by moving between sketching, CAD, fabrication, sensing, and code. Product-design training and controls research bring a systems view of feedback, behavior, and how people and devices respond to one another.

Capabilities

Interaction Prototyping – Physical mockups, haptic and touch concepts, sensors, input/output behaviors
Visual Communication – Hand sketches, Apple Keynote storyboards and animation, CAD renders
Software & Interactive Systems – Python, OpenCV, Arduino, MATLAB/Simulink, sensors, DAQ
Product Development – SolidWorks, Fusion 360, 3D printing, sheet metal, drawings, GD&T, FEA
Systems & Experiments – Feedback, system identification, estimation, test design, design-build-test

Selected Projects

Rapid Interaction Studies: Disc, Orb & Hover Touch — **Aug 2026**

- Developed 3 speculative hardware-interaction concepts through hand sketches, Keynote storyboards, animation, and a rapid physical mockup.
- Built a same-day cardboard Disc prototype to test palm scale, grip, press location, thickness, and display glare before committing to CAD or code.

Autonomous Can-Collection Robot (C.L.E.A.N.): Team Lead — **Apr 2024 - Jun 2024**

- Led a 5-person team through weekly design-build-test cycles, culminating in an autonomous robot that could find, approach, identify, grasp, and lift aluminum cans.
- Trained and tuned an OpenCV cascade classifier on approximately 1,500 custom images; converted the detected can's bounding-box centroid into steering commands on a Raspberry Pi.
- Integrated a custom 3D-printed treaded chassis and 2-DOF worm-gear arm/claw with motor drivers, an Arduino, and ultrasonic and inductive sensors.

Collegiate Electric Race Car (Formula SAE EV): Vehicle Systems Lead — **Aug 2021 - Jun 2025**

- 4-year contributor and Internal VP/VP on an electric race-car design-build-test team; led work across vehicle systems and coordinated teammates through build cycles.
- Designed and built a 600 V accumulator container using an adhesive-first joining strategy; produced CAD, engineering drawings, and structural checks.
- Led braking-system design, manufacturing, and testing; separately won an undergraduate research grant to characterize brake-fluid viscosity and presented the results.

Education

- **University of California, Santa Barbara - Ph.D. Mechanical Engineering** .... **Projected Jun 2029**
  GPA 3.92/4.0 — Control Theory — Year 1 completed; available to take academic leave for full-time work
- **University of California, Santa Barbara - M.S. Mechanical Engineering** .... **Jul 2025**
  GPA 3.90/4.0 — Project: Clogging of fiber suspensions at constrictions
- **University of California, Santa Barbara - B.S. Mechanical Engineering** .... **Jun 2025**
  GPA 3.76/4.0 — Product Design — Capstone: 600 V Vehicle Battery Container

Research Experience

Biological Control Lab, Yeung UCSB - *Research Assistant* **Jan 2025 - Jan 2026**

- Co-developed a control-oriented model of genome-wide transcription coupled to DNA supercoiling dynamics regulated by topoisomerases.
- Supported analysis of stochastic and nonlinear population models (stability/sensitivity analysis) to identify parameters influencing transcription and growth dynamics.
- Co-author on manuscript accepted (Jan 2026); preprint: 10.1101/2025.05.26.655193.

M&M Flow Lab, Sauret UCSB - *Master's Project* **Aug 2024 - Jun 2025**

- Researched clogging dynamics in particle-laden suspensions in constricted flows, relevant to medical and industrial systems.
- Prototyped a vision-based clog detector using a rolling brightness/windowed-visibility metric; used it to support logging and post-analysis of clogging events.
- Presented findings at November 2024 APS Division of Fluid Dynamics meeting.

M&M Flow Lab, Sauret UCSB - *Undergraduate Researcher* **Jan 2023 - Aug 2024**

- Explored cohesive forces in granular materials with 2 PhD student mentors to help understand inter-particle mechanics from the introduction of cohesive forces.
- Designed and built an instrumented annular shear cell; developed an Arduino–MATLAB DAQ + calibration workflow for strain-gauge measurements and repeatable experiments.
- Presented work at Southern California Flow Physics Symposium.

Fluid Energy Science Lab, Luzzatto-Fegiz UCSB - *Undergraduate Researcher* **Jun 2023 - Oct 2023**

- 3D-printed and laser-cut annuli to simulate kites for wind energy applications.
- Data collection for wake model verification in a wind tunnel; made an Arduino-MATLAB DAQ system and app.

Experience - Work

Teaching Assistant - *UCSB: Mech E Academic Student Employee* **Apr 2024 - Present**

- Taught laboratory methods, dynamics, vibrations, and fluids through lab sections, office hours, and course assessment.
- Explained unfamiliar technical ideas, coached bio-inspired design teams, and gave actionable feedback across different learning styles.

Resident Assistant - *UCSB: San Rafael Residence Hall* **Aug 2022 - Jun 2023**

- Managed resident support, conflict resolution, and event planning in a high-traffic dorm environment.

GameMaster - *Diablo Escapes, Walnut Creek* **Jul 2020 - Aug 2022**

- Guided customers through VR escape rooms; troubleshot hardware/software issues and explained unfamiliar tech.
- Helped design puzzles for physical escape rooms under construction.

Honors

- Undergraduate Engineering Honors Program **Sept 2023 - Jun 2024**
- Dean's Honors (Engineering), 5 quarters
- 2 Undergraduate Research Awards ($3750 and $2500) **2024 and 2025**

Last updated: August 10, 2026